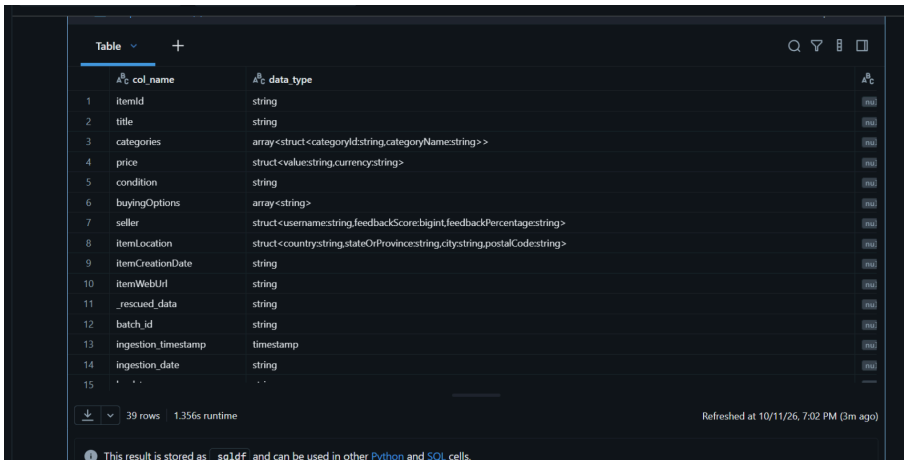
Explicit schema proof

1. Bronze layer

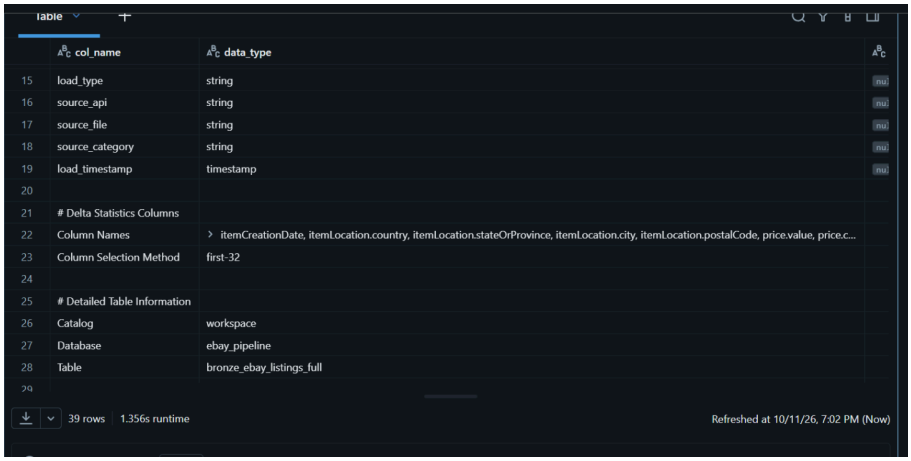


Table

	col_name	data_type
1	itemId	string
2	title	string
3	categories	array<struct<categoryId:string,categoryName:string>>
4	price	struct<value:string,currency:string>
5	condition	string
6	buyingOptions	array<string>
7	seller	struct<username:string,feedbackScore:bigint,feedbackPercentage:string>
8	itemLocation	struct<country:string,stateOrProvince:string,city:string,postalCode:string>
9	itemCreationDate	string
10	itemWebUrl	string
11	_rescued_data	string
12	batch_id	string
13	ingestion_timestamp	timestamp
14	ingestion_date	string
15		

39 rows | 1.356s runtime — Refreshed at 10/11/26, 7:02 PM (3m ago)

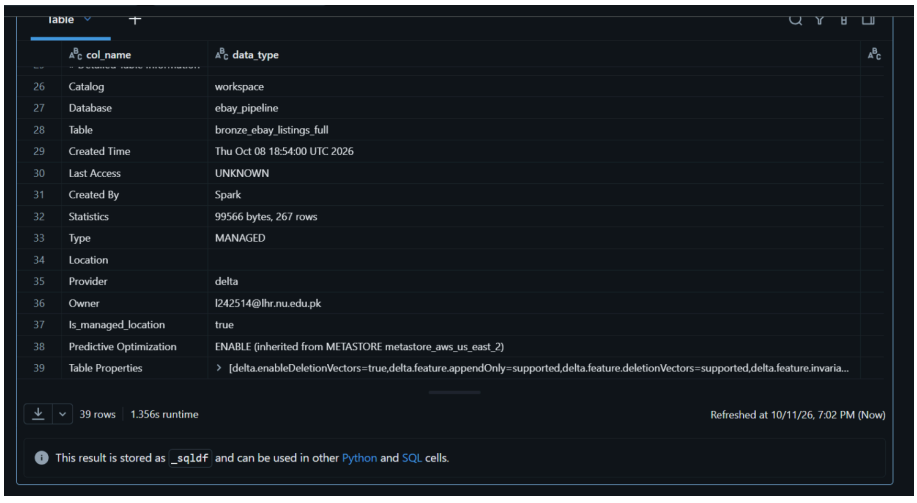
This result is stored as `_sqldf` and can be used in other Python and SQL cells.



Table

	col_name	data_type
15	load_type	string
16	source_api	string
17	source_file	string
18	source_category	string
19	load_timestamp	timestamp
20		
21	# Delta Statistics Columns	
22	Column Names	itemCreationDate, itemLocation.country, itemLocation.stateOrProvince, itemLocation.city, itemLocation.postalCode, price.value, price.c...
23	Column Selection Method	first-32
24		
25	# Detailed Table Information	
26	Catalog	workspace
27	Database	ebay_pipeline
28	Table	bronze_ebay_listings_full

39 rows | 1.356s runtime — Refreshed at 10/11/26, 7:02 PM (Now)



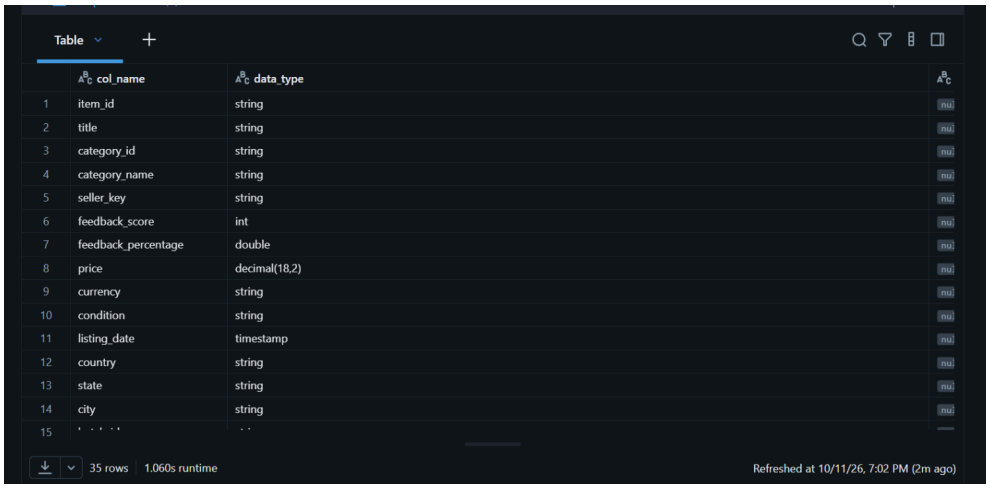
Table

	col_name	data_type
26	Catalog	workspace
27	Database	ebay_pipeline
28	Table	bronze_ebay_listings_full
29	Created Time	Thu Oct 08 18:54:00 UTC 2026
30	Last Access	UNKNOWN
31	Created By	Spark
32	Statistics	99566 bytes, 267 rows
33	Type	MANAGED
34	Location	
35	Provider	delta
36	Owner	l242514@lhr.nu.edu.pk
37	Is_managed_location	true
38	Predictive Optimization	ENABLE (inherited from METASTORE metastore_aws_us_east_2)
39	Table Properties	[delta.enableDeletionVectors=true,delta.feature.appendOnly=supported,delta.feature.deletionVectors=supported,delta.feature.invaria...

39 rows | 1.356s runtime — Refreshed at 10/11/26, 7:02 PM (Now)

This result is stored as `_sqldf` and can be used in other Python and SQL cells.

Silver

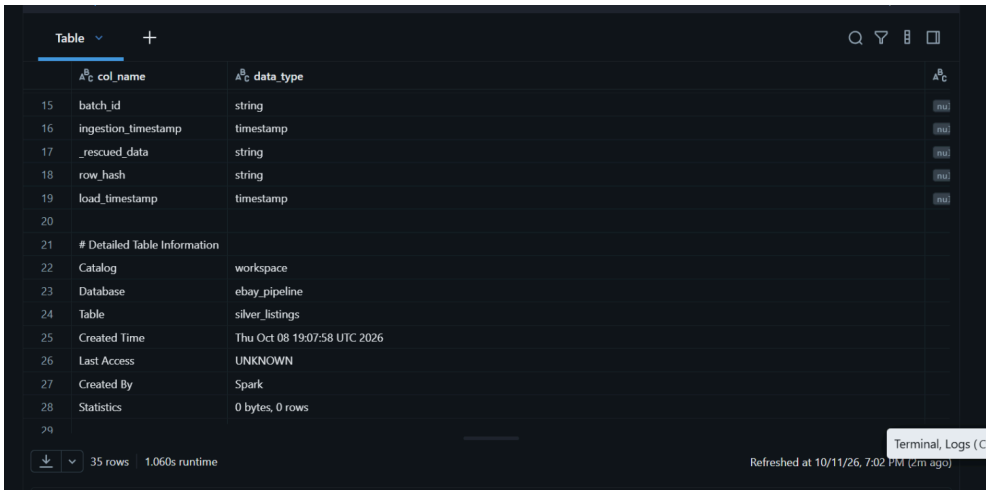


Table

	col_name	data_type
1	item_id	string
2	title	string
3	category_id	string
4	category_name	string
5	seller_key	string
6	feedback_score	int
7	feedback_percentage	double
8	price	decimal(18,2)
9	currency	string
10	condition	string
11	listing_date	timestamp
12	country	string
13	state	string
14	city	string
15		

35 rows | 1.060s runtime

Refreshed at 10/11/26, 7:02 PM (2m ago)



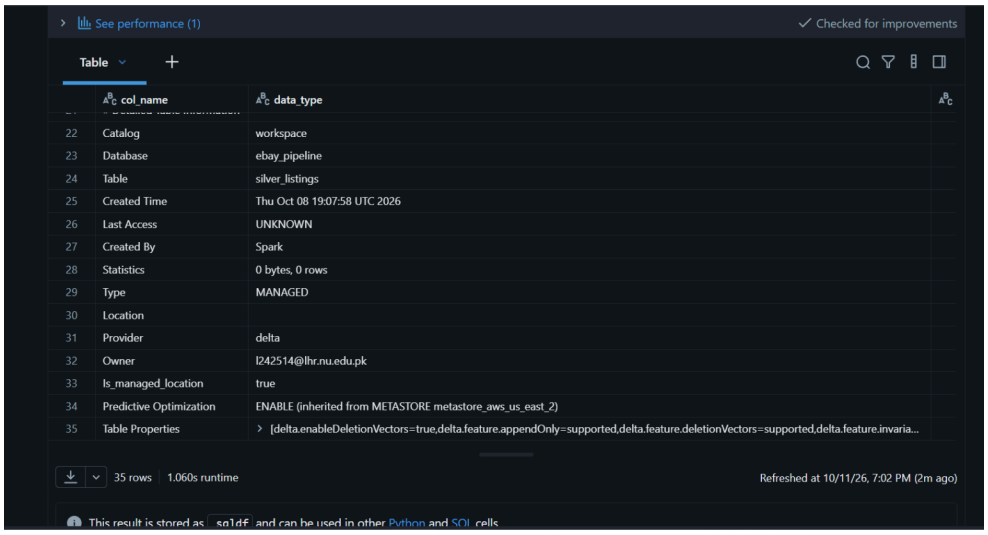
Table

	col_name	data_type
15	batch_id	string
16	ingestion_timestamp	timestamp
17	_rescued_data	string
18	row_hash	string
19	load_timestamp	timestamp
20		
21	# Detailed Table Information	
22	Catalog	workspace
23	Database	ebay_pipeline
24	Table	silver_listings
25	Created Time	Thu Oct 08 19:07:58 UTC 2026
26	Last Access	UNKNOWN
27	Created By	Spark
28	Statistics	0 bytes, 0 rows
29		

Terminal, Logs (C

35 rows | 1.060s runtime

Refreshed at 10/11/26, 7:02 PM (2m ago)



See performance (1)

Checked for improvements

Table

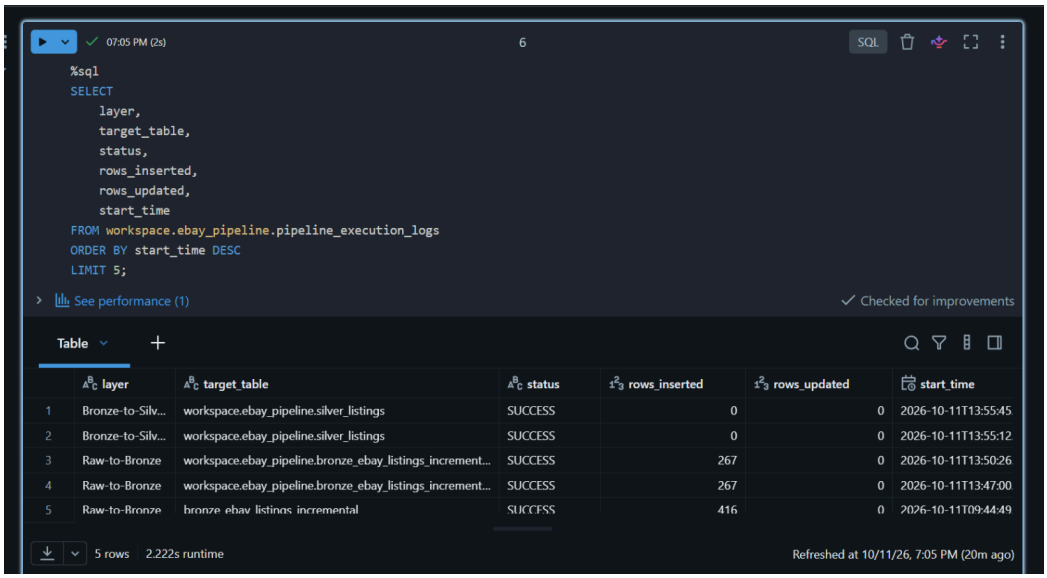
	col_name	data_type
22	Catalog	workspace
23	Database	ebay_pipeline
24	Table	silver_listings
25	Created Time	Thu Oct 08 19:07:58 UTC 2026
26	Last Access	UNKNOWN
27	Created By	Spark
28	Statistics	0 bytes, 0 rows
29	Type	MANAGED
30	Location	
31	Provider	delta
32	Owner	l242514@lhr.nu.edu.pk
33	Is_managed_location	true
34	Predictive Optimization	ENABLE (inherited from METASTORE metastore_aws_us_east_2)
35	Table Properties	> [delta.enableDeletionVectors=true,delta.feature.appendOnly=supported,delta.feature.deletionVectors=supported,delta.feature.invaria...

35 rows | 1.060s runtime

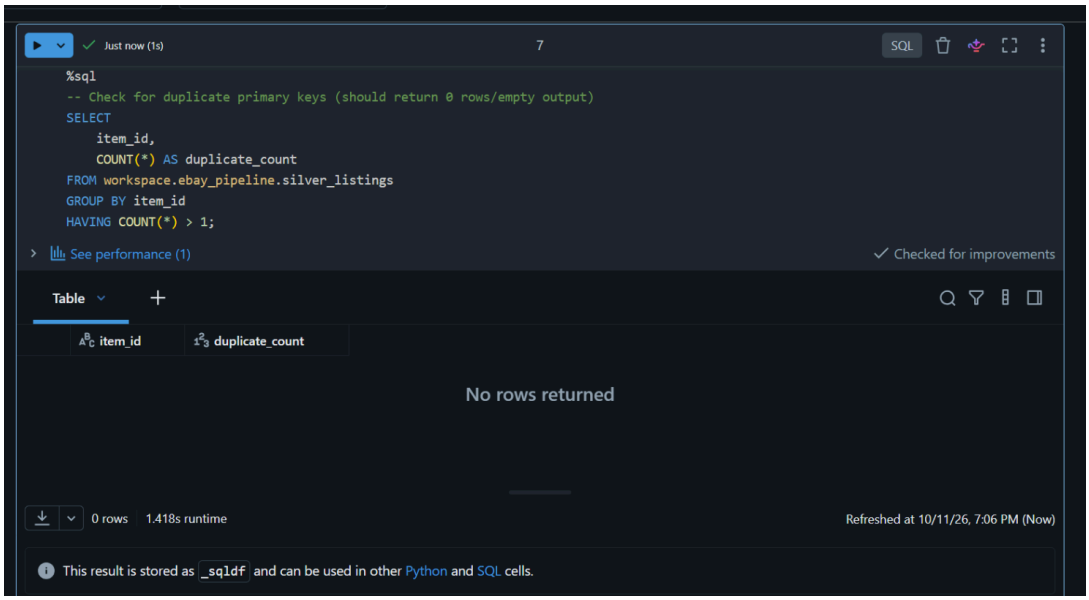
Refreshed at 10/11/26, 7:02 PM (2m ago)

This result is stored as `sqldf` and can be used in other Python and SQL cells

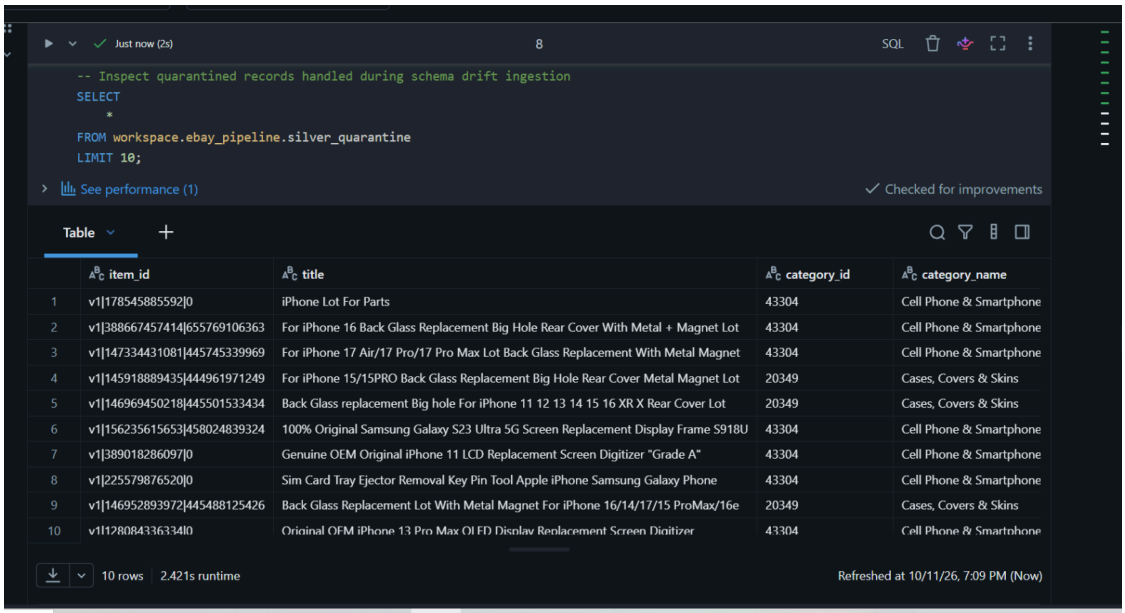
Idempotency



Primary key uniqueness verification



Schema drift and quarantine proof



Just now (2s) 8 SQL

```
-- Inspect quarantined records handled during schema drift ingestion
SELECT
    *
FROM workspace.ebay_pipeline.silver_quarantine
LIMIT 10;
```

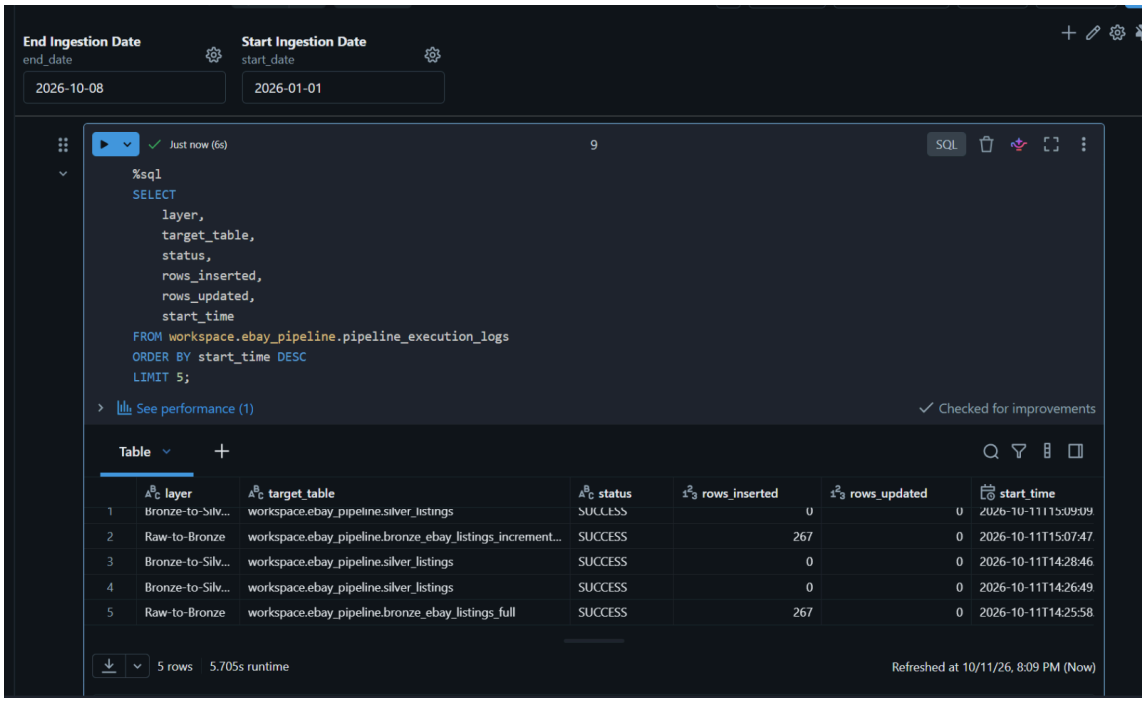
See performance (1) Checked for improvements

Table

	item_id	title	category_id	category_name
1	v1|178545885592|0	iPhone Lot For Parts	43304	Cell Phone & Smartphone
2	v1|388667457414|655769106363	For iPhone 16 Back Glass Replacement Big Hole Rear Cover With Metal + Magnet Lot	43304	Cell Phone & Smartphone
3	v1|147334431081|445745339969	For iPhone 17 Air/17 Pro/17 Pro Max Lot Back Glass Replacement With Metal Magnet	43304	Cell Phone & Smartphone
4	v1|145918889435|444961971249	For iPhone 15/15PRO Back Glass Replacement Big Hole Rear Cover Metal Magnet Lot	20349	Cases, Covers & Skins
5	v1|146969450218|445501533434	Back Glass replacement Big hole For iPhone 11 12 13 14 15 16 XR X Rear Cover Lot	20349	Cases, Covers & Skins
6	v1|156235615653|458024839324	100% Original Samsung Galaxy S23 Ultra 5G Screen Replacement Display Frame S918U	43304	Cell Phone & Smartphone
7	v1|389018286097|0	Genuine OEM Original iPhone 11 LCD Replacement Screen Digitizer "Grade A"	43304	Cell Phone & Smartphone
8	v1|2255798765 20|0	Sim Card Tray Ejector Removal Key Pin Tool Apple iPhone Samsung Galaxy Phone	43304	Cell Phone & Smartphone
9	v1|146952893972|445488125426	Back Glass Replacement Lot With Metal Magnet For iPhone 16/14/17/15 ProMax/16e	20349	Cases, Covers & Skins
10	v1|128084336334|0	Original OEM iPhone 13 Pro Max OLED Display Replacement Screen Digitizer	43304	Cell Phone & Smartphone

10 rows 2.421s runtime Refreshed at 10/11/26, 7:09 PM (Now)

Historical Backfill Execution and Operational Logs



End Ingestion Date
end_date
2026-10-08

Start Ingestion Date
start_date
2026-01-01

Just now (6s) 9 SQL

```
%sql
SELECT
    layer,
    target_table,
    status,
    rows_inserted,
    rows_updated,
    start_time
FROM workspace.ebay_pipeline.pipeline_execution_logs
ORDER BY start_time DESC
LIMIT 5;
```

See performance (1) Checked for improvements

Table

	layer	target_table	status	rows_inserted	rows_updated	start_time
1	Bronze-to-Silv...	workspace.ebay_pipeline.silver_listings	SUCCESS	0	0	2026-10-11T15:09:09
2	Raw-to-Bronze	workspace.ebay_pipeline.bronze_ebay_listings_increment...	SUCCESS	267	0	2026-10-11T15:07:47
3	Bronze-to-Silv...	workspace.ebay_pipeline.silver_listings	SUCCESS	0	0	2026-10-11T14:28:46
4	Bronze-to-Silv...	workspace.ebay_pipeline.silver_listings	SUCCESS	0	0	2026-10-11T14:26:49
5	Raw-to-Bronze	workspace.ebay_pipeline.bronze_ebay_listings_full	SUCCESS	267	0	2026-10-11T14:25:58

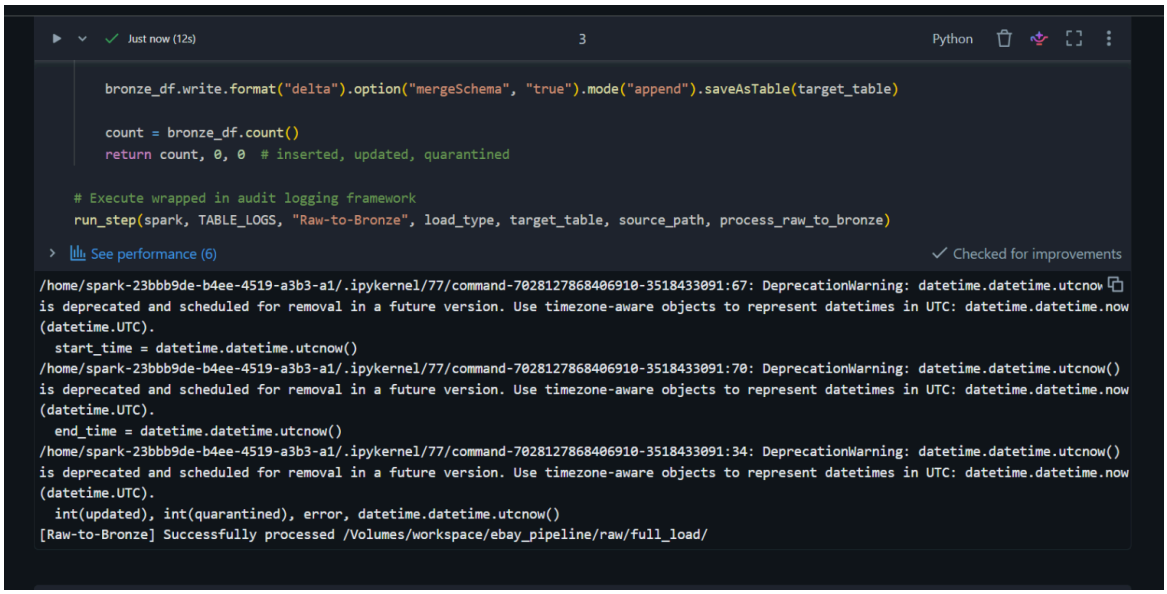
5 rows 5.705s runtime Refreshed at 10/11/26, 8:09 PM (Now)

Running of all notebooks

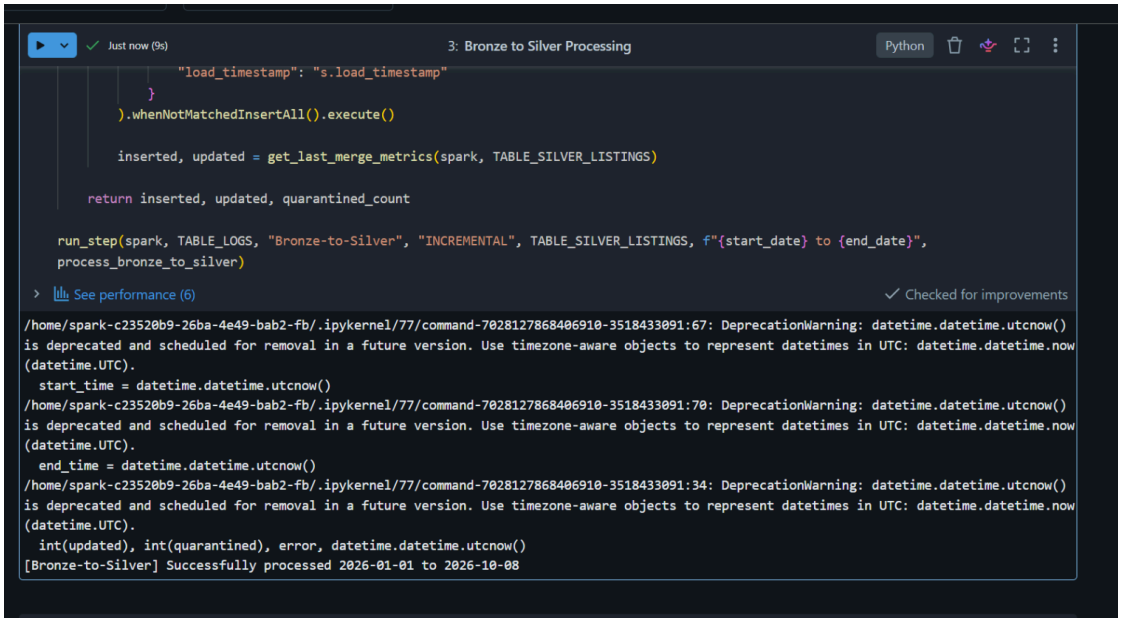
1. config



2. Bronze



3. Silver


Just now (9s)
3: Bronze to Silver Processing
Python
"load_timestamp": "s.load_timestamp"
}
).whenNotMatchedInsertAll().execute()
inserted, updated = get_last_merge_metrics(spark, TABLE_SILVER_LISTINGS)
return inserted, updated, quarantined_count
run_step(spark, TABLE_LOGS, "Bronze-to-Silver", "INCREMENTAL", TABLE_SILVER_LISTINGS, f"{start_date} to {end_date}",
process_bronze_to_silver)
See performance (6)
Checked for improvements
/home/spark-c23520b9-26ba-4e49-bab2-fb/.ipykernel/77/command-7028127868406910-3518433091:67: DeprecationWarning: datetime.datetime.utcnow()
is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now
(datetime.UTC).
start_time = datetime.datetime.utcnow()
/home/spark-c23520b9-26ba-4e49-bab2-fb/.ipykernel/77/command-7028127868406910-3518433091:70: DeprecationWarning: datetime.datetime.utcnow()
is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now
(datetime.UTC).
end_time = datetime.datetime.utcnow()
/home/spark-c23520b9-26ba-4e49-bab2-fb/.ipykernel/77/command-7028127868406910-3518433091:34: DeprecationWarning: datetime.datetime.utcnow()
is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now
(datetime.UTC).
int(updated), int(quarantined), error, datetime.datetime.utcnow()
[Bronze-to-Silver] Successfully processed 2026-01-01 to 2026-10-08